

# Shell

PENG

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## Contents

|          |                             |          |
|----------|-----------------------------|----------|
| <b>1</b> | <b>Introduction</b>         | <b>1</b> |
| <b>2</b> | <b>Shell Syntax</b>         | <b>1</b> |
| 2.1      | Quoting . . . . .           | 1        |
| 2.2      | Pipelines . . . . .         | 2        |
| 2.3      | Lists of Commands . . . . . | 2        |
| 2.4      | Construct . . . . .         | 2        |
| 2.4.1    | Looping . . . . .           | 2        |
| 2.4.2    | Conditional . . . . .       | 2        |
| 2.4.3    | Grouping . . . . .          | 2        |
| 2.5      | Functions . . . . .         | 2        |
| 2.6      | Parameters . . . . .        | 3        |
| 2.7      | Expansion . . . . .         | 3        |
| 2.8      | Redirection . . . . .       | 3        |
| <b>3</b> | <b>Array</b>                | <b>3</b> |
| <b>4</b> | <b>sed</b>                  | <b>4</b> |
| <b>5</b> | <b>awk</b>                  | <b>4</b> |
| <b>6</b> | <b>grep</b>                 | <b>4</b> |
| <b>7</b> | <b>find</b>                 | <b>4</b> |
| <b>8</b> | <b>Windows Batch</b>        | <b>4</b> |
| <b>A</b> | <b>Tools</b>                | <b>4</b> |

## 1 Introduction

<https://www.gnu.org/software/bash/manual/bash.html>

## 2 Shell Syntax

### 2.1 Quoting

**Definition 2.1** (quoting). *Quoting is used to remove the special meaning of certain characters or words to the shell.*

```
echo -e "\t"
# \n : newline
# \t : horizontal tab
# \r : carriage return
```

## 2.2 Pipelines

**Definition 2.2** (pipeline). *cmd1 output*  $\rightarrow$  *cmd2 input*

```
cmd1 | cmd2      # stdout
cmd1 |& cmd2     # stderr to stdout
cmd1 2>&1 | cmd2 # stderr to stdout
```

## 2.3 Lists of Commands

**Definition 2.3** (list).

```
cmd1 && cmd2 # And, cmd1 true then cmd2
cmd1 || cmd2 # Or, cmd1 false then cmd2
cmd1; cmd2   # or terminated by newline
cmd1 & cmd2  # cmd1 async
```

## 2.4 Construct

### 2.4.1 Looping

**Definition 2.4** (looping).

```
for item in item1 item2 ...; do
    cmd
done
```

### 2.4.2 Conditional

**Definition 2.5** (conditional).

```
if [[ cond1 ]]; then
    cmd1
elif [[ cond2 ]]; then
    cmd2
else
    cmd3
fi
```

```
# cond:
# ==
# !=
# =~ : unquoted regex
```

### 2.4.3 Grouping

**Definition 2.6** (grouping).

```
( expr ) # subshell env, def var vanish when exit
{ expr; } # ; or newline is required
```

## 2.5 Functions

**Definition 2.7** (function).

```
func () {
    local var = $1
}
```

## 2.6 Parameters

**Definition 2.8** (parameters).

- `${N}` : *position param*
- `$@` : *expands to all pos params, \$1 \$2 ...*
- `$0` : *file or shell name*

## 2.7 Expansion

**Definition 2.9** (brace expansion). *Generate strings sharing a common prefix and suffix.*

```
echo file{1,2} # file1 file2
```

**Definition 2.10** (tidle expansion).

```
~ # $HOME
```

**Definition 2.11** (parameter expansion).

```
${param}
```

```
${param:-default}
```

```
${array[@]} # member of array
```

```
${!name[@]} # list of array indices
```

**Definition 2.12** (command substitution).

```
$(cmd)
```

```
`cmd`
```

**Definition 2.13** (arithmetic expansion).

```
$((expr)) # integer-only
```

**Definition 2.14** (filename expansion).

```
* # any string
```

```
? # single char
```

```
[] # char enclosed between []
```

```
*()
```

```
?()
```

```
+()
```

```
!(pattern list) # match except given patterns
```

## 2.8 Redirection

**Definition 2.15** (redirection).

```
ls > file 2>&1 # stdout and stderr to file
```

```
ls &> file # or
```

```
pwd >> file 2>&1 # append stdout and stderr to file
```

```
pwd &>> file # or
```

## 3 Array

**Definition 3.1** (array).

```
array=(v1 v2 ...)
```

```
for i in ${!array[@]}; do
```

```
    echo $array[$i]
```

```
done
```

## 4 sed

## 5 awk

<https://www.gnu.org/software/gawk/manual/gawk.html>

**Definition 5.1** (awk). *The basic function of awk is to search files for lines that contain certain patterns.*

**Example 5.2.**

```
# % @fig{png}
mapfile -t array < <(awk -F'[{,}]' '/^% @fig\{/ { print $2 }' file)

# @book{BATCHELOR_2000_AN_INTRODUCTION_TO_FLUID_DYNAMICS,
awk -F'[{,}]' '/@book\{/ {
    print "$1: " $1
    print "$2: " $2
    print "$3: " $3
}' file
```

## 6 grep

**Definition 6.1** (grep). *Search input file for matches to the patterns.*

## 7 find

## 8 Windows Batch

## A Tools

```
#!/bin/bash

# Convert TikZ PDF to PNG
# pdf2png PDF PAGE PNG [DPI]
pdf2png() {
    local PDF=$1
    local PAGE=$2
    local PNG=$3
    local DPI=${4:-300}
    echo "Convert $PDF page $PAGE to $PNG"
    magick -density "$DPI" -units pixelsperinch \
        "$PDF[$PAGE]" \
        -alpha remove -background white -flatten \
        -trim +repage \
        -quality 100 \
        "$PNG"
}

#!/bin/bash

# Compress PDF
# pdfcompress INPUT OUTPUT
pdfcompress() {
    local INPUT=$1
    local OUTPUT=$2
    ghostscript -sDEVICE=pdfwrite -dPDFSETTINGS=/prepress -dNOPAUSE -dQUIET -dBATCH \
        -sOUTPUTFile="$OUTPUT" "$INPUT"
}
```